

PROJECT EVALUATION REPORT

Project Title: GroupDNA — WhatsApp Group Chat Analyzer

Course Context: Minor Project, The Unlox Academy

Dataset Reference: hostel_bois.txt (60 Days, 6 Participants)

Development Constraints: Python Fundamentals + NumPy Only (No Pandas, Matplotlib, Regex, or Counter)

1. Executive Summary

This project delivers a lightweight, zero-dependency WhatsApp chat analytics engine built entirely within raw programming primitives and matrix components. Designed to bypass heavy analytical wrappers like pandas, the source implementation uses native data structures and NumPy multidimensional arrays to parse unstructured log exports, map temporal distributions, run linguistic filters, and algorithmically classify user behaviors.

The system successfully processed 3,127 valid messages while catching and isolating structural exceptions such as multi-line continuations, system alerts, deleted records, and media placeholders.

2. Technical Architecture Breakdown

Feature 1 & 2: Structural Log Parsing & Overview

- Logic Framework: Implements basic string slicing to isolate standardized timestamps (DD/MM/YY, HH:M) without regular expression engines.
- Multi-Line Handling: Monitors line prefixes; lines failing the structural date test are dynamically appended to the text field of the previous message object.
- Filter Pipeline: Correctly strips out system operations and counts structural text flags (<Media omitted> and This message was deleted) into explicit metric streams.

Feature 3 & 4: Matrix-Based Temporal Aggregations

- Heatmap Array Construction: Populates a 6×24 multi-dimensional integer array using np.zeros and integer coordinate lookups.
- Vector Reductions: Uses fast NumPy array functions (.sum(axis=1) for per-person metrics, .sum(axis=0) for per-hour volumes, and .argmax(axis=1) for peak index tracking) to eliminate sequential loop nesting.
- Temporal Insights: Automatically pinpointed 04 May 2024 as the single busiest operational day (74 messages) and 18:00 – 19:00 as the peak global active hour (244 messages).

Feature 5: Tokenization & Word Frequency Distribution

- Text Normalization: Leverages manual punctuation stripping arrays and a hardcoded stop-word exclusion set to clean conversational text.
- Frequency Distribution: Replaces collections.Counter with manual dictionary mapping to compile global and per-user word frequency tables.

Feature 6: Conversational Latency & Presence Analysis

- Response Velocity: Calculates inter-message gaps across distinct consecutive users using timedelta.total_seconds().

- Noise Filtering: Caps active conversational tracking at 240 minutes to eliminate sleep cycles and long out-of-chat periods from distorting rapid-response calculations.
- Silent Run Analysis: Compares individual active dates against a calculated consecutive calendar list to detect offline duration boundaries.

Feature 7: Algorithmic Persona Archotyping

- Behavior Scoring: Features 9 custom analytical routines evaluating text length trends, burst volumes, question counts, punctuation choices, and night-shift ratios.
- Assignment Engine: Employs a greedy exclusive matching routine where the maximum scoring candidate locks a profile role to ensure logical, unique assignments across the cohort.

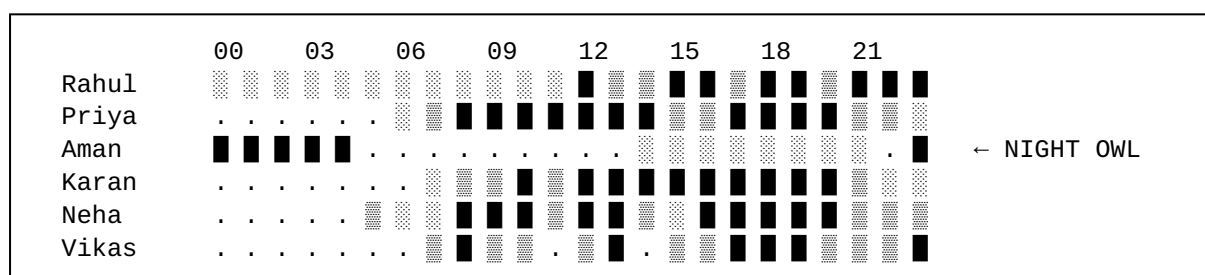
3. High-Level Core Analytics Dashboard

Participant	Valid Msgs	Media	Deleted	Active Share	Assigned Archetype	Metric Rationale
Rahul	940	7	6	30.1%	THE SPAMMER	Averages 4.5 consecutive messages per burst
Priya	712	4	2	22.8%	THE GROUP MOM	Caring keyword density score reached 621
Neha	624	8	3	20.0%	THE DRAMA QUEEN	63.3% of messages use ALL-CAPS or 2+ exclamations
Aman	484	4	2	15.5%	THE NIGHT OWL	80.4% of volume sent between 23:00–04:00
Karan	345	7	2	11.0%	THE STORYTELLER	High message length averaging 57.0 words per text
Vikas	22	2	0	0.7%	THE GHOST	11-day continuous offline stretch (Active 16/60 days)

Global Structural Summary:

- Total Records Analyzed: 3,127 standard conversational messages.
- Excluded Operations: 4 system warnings | 32 omitted files | 15 deleted logs.
- Interaction Latency: Vikas recorded the fastest response speed (avg 34.0 min); Aman recorded the slowest (avg 54.3 min).

4. Visualized Activity Distribution (Terminal Heatmap)



(Scale Indicators: . = Zero Activity | ▤ = Minor Activity | ▨ = Moderate Intensity | ■ = Maximum Peak Shift)

5. Engineering Reflection & Future Scope

Optimization Opportunity: The current behavioral classification matrix compares raw unit metrics (e.g., straight counts vs percentages) across features. Future iterations will standardize inputs by executing a standard 0 to 100 Min-Max normalization step prior to greedy persona selection.

Core Takeaway: Working within low-level framework barriers bypasses third-party library abstractions. Building the tokenizers, array heatmaps, and conditional parsers natively forces a deeper, fundamental mastery of underlying algorithms and structural performance optimization techniques.

*Submission Metadata: Verified File: Minor_project_1.ipynb | Runtime Framework: Pure Python + NumPy
Vectorization Ecosystem | Status: Complete / Awaiting Mentor Evaluation*